

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Krish Wadhwa

Enrollment No. 23103060

Jaypee Institute of Information Technology, Noida
Test-2 Examination, ODD 2023
B. Tech, 1st Semester

Course Title: Software Development Fundamentals-1
Course Code: 15B11CI111

Maximum Time: 1 Hour
Maximum Marks: 20

COs	Course Outcomes	LEVEL
C109.1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithm and flowcharts.	Understand (L2)
C109.2	Explain basics of C programming concepts to make decision for solving problems.	Understand (L2)
C109.3	Demonstrate and contrast different methods for writing modular programs in C	Understand (L2)
C109.4	Use various C programming constructs to implement iteration, and recursion.	Apply (L3)
C109.5	Apply and implement arrays, pointers, structures and file handling for solving real world problems.	Apply (L3)

Attempt all questions:

Q1. Write a function in C that can take an integer as an argument and returns the sum of factors it has. Call this function via the main function and print the sum of factors in the main function.
Example:

sumOfFactors(10) will return 18 (i.e. $1+2+5+10=18$)

[C109.3 (Understand L2), 4 marks]

Q2. Write a program in C that should print the number with the highest frequency in a given array. You are given an array of non-negative integers where each integer is less than 10, and your task is to find the number that appears most frequently in the array. If there are multiple numbers with the same highest frequency, then print the smallest number whose frequency is highest.

Example 1: Given the input array [2,3,2,4,2,5,2], the program should print 2 because 2 appears with the highest frequency in the array.

Example 2: Given the input array [2,3,5,4,3,5,5,3], the program should print 3 because 3 (smallest number) appears with the highest frequency in the array.

[C109.5 (Apply L3), 4 marks]

Q3. Write a program in C that performs the following tasks:

- Ask the user to input a square matrix A.
- Detects the maximum element in the upper triangular matrix of A (including the diagonal elements).
- Detects the minimum element in the lower triangular matrix of A (including the diagonal elements).
- If the sum of these two elements (max in upper triangular and min in lower triangular matrix) is equal to the sum of elements in the leading diagonal (from top left to bottom right) of the original matrix then print: "Traceable" else print "Non-Traceable".

[C109.5 (Apply L3), 1 + 1.5 + 1.5 + 2 Marks]

Test cases are provided on the next page.

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Jaypee Institute of Information Technology, Noida

Test 1 Examination, ODD 2023

B.Tech(CSE/IT/ECE)/Dual, I Semester

Course Title: Software Development Fundamentals-1

Time: 1 hour

Course Code: 15B11CI111

Maximum Marks: 20

COs	Course Outcomes
C109.1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithm and flowcharts:
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C109.4	Use various C programming constructs to implement iteration, and recursion
C109.5	Apply and implement arrays, pointers, structures and file handling for solving real world problems

Attempt all questions

Q1. Four people (Alex, Bob, Carol, and Dave) come across a bridge at night. The bridge can only hold two people at a time, and they only have one flashlight (crossing the bridge without flashlight is impossible). The four people walk at different speeds: Alex takes 1 minute to cross the bridge, Bob takes 2 minutes, Carol takes 5 minutes, and Dave takes 10 minutes. When two people cross the bridge together, they must go at the slower person's pace. They have only 17 minutes before a train passes under the bridge, and they need to get to the other side before that happens. Your task is to find the shortest time in which all four people can safely cross the bridge. Explain your steps. [C109.1 (Understanding), 2 Marks]

Q2. You have been asked to design an automated face recognition system. However, the exact requirements are not yet clear. Can we use waterfall SDLC model for this particular system? Justify your answer. [C109.1 (Understanding), 1 Marks]

Q3. Write a program to input a positive integer from user. Then exchange its i^{th} and j^{th} digits. The value of i and j should be given as input by the user. Please note all the digits in the number are supposed to be non-zero.

Example: $N = 3246159$

$i = 2, j = 5$

i^{th} digit is 5 and j^{th} digit is 4

Exchanged number is 3256149

[C109.4 (Applying), 5 Marks]

Test Case 1:

Enter the size of the square matrix A: 3
Enter the element of the square matrix A (row by row):
1 2 3
4 5 6
7 8 0
Maximum element in the upper triangular matrix: 6
Minimum element of the lower triangular matrix: 0
Sum of max and min elements: 6
Sum of elements in the leading diagonal: 6
Traceable

Test Case 2:

Enter the size of the square matrix A: 3
Enter the elements of the square matrix A (row by row):
1 2 9
4 5 6
7 8 5
Maximum element in upper triangular matrix: 9
Minimum element in lower triangular matrix : 1
Sum of max and min elements: 10
Sum of elements in the leading diagonal: 11
Non- Traceable

Q4. Write a C program that takes a string as input and reverses each word in the string. A word is defined as a sequence of non- space characters. You should maintain the order of words in the original string, but reverse the characters within each word. For example, if the input string is "Hello World", your program should modify it to become "olleH dlroW".

Note:

String should be accessed via pointer variable

You can assume that the input string contains only printable ASCII characters and is null-terminated.

[C109.5 (Apply L3), 4 Marks]

Q5. Write the output of the following code with proper explanation:

```
#include<stdio.h>
#include<string.h>
int main()
{
    char p[20];
    char *s = "string";
    int length = strlen(s);
    int i;
    for (i=0; i<length; i++)
        p[i] = s[length-i];
    printf("%s",p);
    return 0;
}
```

[C109.5 (Apply L3), 2 Marks]

pos=0;
p[i] (ch== ' ')

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Jaypee Institute of Information Technology, Noida

Test 1 Examination, ODD 2023

B.Tech(CSE/IT/ECE)/Dual, I Semester

Course Title: Software Development Fundamentals-1 Time: 1 hour

Course Code: 15B11CI111

Maximum Marks: 20

COs	Course Outcomes
C109.1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithm and flowcharts.
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Attempt all questions

Q1. Four people (Alex, Bob, Carol, and Dave) come across a bridge at night. The bridge can only hold two people at a time, and they only have one flashlight (crossing the bridge without flashlight is impossible). The four people walk at different speeds: Alex takes 1 minute to cross the bridge, Bob takes 2 minutes, Carol takes 5 minutes, and Dave takes 10 minutes. When two people cross the bridge together, they must go at the slower person's pace. They have only 17 minutes before a train passes under the bridge, and they need to get to the other side before that happens. Your task is to find the shortest time in which all four people can safely cross the bridge. Explain your steps. [C109.1 (Understanding), 2 Marks]

Q2. You have been asked to design an automated face recognition system. However, the exact requirements are not yet clear. Can we use waterfall SDLC model for this particular system? Justify your answer. [C109.1 (Understanding), 1 Marks]

Q3. Write a program to input a positive integer from user. Then exchange its i^{th} and j^{th} digits. The value of i and j should be given as input by the user. Please note all the digits in the number are supposed to be non-zero.

Example: $N = 3246159$

$i = 2, j = 5$

i^{th} digit is 5 and j^{th} digit is 4

Exchanged number is 3256149

[C109.4 (Applying), 5 Marks]

Q4. Write a C program to automate the process of apparel label generation for Men's trousers. The program should take inputs of standard Hip and Length in inches. The program should identify the standard apparel Label Size based on the table using Switch-case control as given below.

Hip (in inches)	Length (in inches)	Label Size
30	38	LARGE
32	39	XL
34	40	XXL

Finally, the output should display the appropriate label of Hip, Length, and Label Size. Sample label format generated is given as below.

Hip : 32
Length : 39
Label Size : XL

For any other input display the message "Wrong Input".

[C109.2 (Understanding), 3 Marks]

Q5. You are asked to draw a flowchart that determines a person's income tax based on their income and other criteria. The tax calculation is based on the following rules:

Income	Tax
Less than 700000	0
Greater than 700000 and less than equal to 1000000	10%
Greater than 1000000 and less than equal to 1500000	30000 + 20% of amount exceeding
Greater than 1500000	130000 + 30% of amount exceeding

Additionally, if the total tax of an individual exceeds 30,000 then an additional 5% education cess tax is applied on their tax.

[C109.1 (Understanding), 3 Marks]

Q6. Write a C program to print a butterfly pattern based on the height of the butterfly provided by the user. Your program must ensure that the height entered by the user must be an even positive integer.

Enter height: 10

[C109.4 (Applying), 6 Marks]

```

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*

```

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name Krish Wadhwa

Enrollment No. 23103060

Jaypee Institute of Information Technology, Noida

End-Term Examination, 2023

B.Tech (CSE/IT/ECE)/INTG(CSE/ECE), 1st Semester

Course Title: Software Development Fundamentals - I
Course Code: 15B11CI111

Maximum Time: 2 hr.

Maximum Marks: 35

Course Outcomes:

CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts.
CO2	Explain basics of C programming concepts to make decisions for solving problems.
CO3	Demonstrate and contrast different methods for writing modular programs in C.
CO4	Use various C programming constructs to implement iteration and recursion.
CO5	Apply and implement arrays, pointers, structures and file handling for solving real world problems.

Note: Attempt all questions

Q1. Draw a flowchart for finding all the roots of 'N' different quadratic equations (where N is provided by the user) of the form $ax^2+bx+c=0$, where a, b and c are real constants. Formula for calculating the roots of the quadratic equation is as follows:

$$\text{Roots} = \frac{-b \pm \sqrt{D}}{2a}$$

$$\text{where } D = (b * b) - (4 * a * c)$$

Your flowcharts should additionally show whether the roots will be "real and equal" or "real and unequal" or "imaginary".

[C01, 4 Marks, Understand]

Q2. Given an unsorted array Arr of N numbers (any combination of positive and negative). Your task is to create an array of alternate positive and negative numbers without changing the relative order of positive and negative numbers. It is ensured that the array has at least one positive number.

Note: Resultant array should start with a positive number and zero should be considered as a positive element.

Input: N = 9

Arr[] = {9,4,-2,-1,5,0,-5,-3,2}

Output: 9, -2, 4, -1, 5, -5, 0, -3, 2

Input: N = 10

Arr[] = {-5,-2,15,2,4,7,1,8,0,-8}

Output: 15, -5, 2, -2, 4, -8, 7, 1, 8, 0

[C02, 6 Marks, Understand]

Q3. Write a program to validate an IPv4 address. According to Wikipedia, IPv4 address are canonically represented in dot-decimal notation, which consists of four decimal numbers, each ranging from 0 to 255, separated by dots, i.e. 172.16.254.1

A valid IPv4 address is of the form $x_1.x_2.x_3.x_4$ where $0 \leq (x_1, x_2, x_3, x_4) \leq 255$. Thus we can write the generalized form of an IPv4 address (0-255). (0-255). (0-255). (0-255). Your task is to write a function **isValid()** which returns 1 if the given IPv4 address is valid, else return 0. The function takes the IPv4 address as the only argument in the form of string.

Note: Here we are considering numbers only from 0-255 and only additional leading zeros will be considered invalid. You are not supposed to use **strtoken()** function.

P.T.O

Sample Input: IPv4 address = 222.111.111.111

Output: 1

Sample Input: IPv4 address = 5555..555

Output: 0

Sample Input: IPv4 address = 02.111.111.111

Output: 0

[C03, 6 Marks, Understand]

Q4. Write a recursive function to print the Aliquot Sequence of the given number 'n'. Aliquot Sequence of a number start with itself, remaining terms of a sequence are sum of proper divisors of immediate previous term. For example, Aliquot Sequence for 10 is 10,8,7,1,0. The sequence may repeat. For example, for 6, we have an infinite sequence of all 6s. In such cases we print the repeating number and stop.

Explanation:

Input: n = 10

Output: 10 8 7 1 0

Sum of proper divisors of 10 is $5+2+1 = 8$

Sum of proper divisors of 8 is $4+2+1 = 7$

Sum of proper divisors of 7 is 1

Sum of proper divisors of 1 is 0

Note: There is no proper divisor of 1

Test Cases:

Input n: 6

Output = 6 repeats with 6

Input n: 12

Output = 12 16 15 9 4 3 1 0

[C04, 6 Marks, Apply]

Q5. A testing agency is conducting an examination for class XIIth students of two different streams. Every student has name (string) and roll-no (integer). Additionally, a science stream's student has marks in physics, chemistry and mathematics while an arts student has the marks in English and regional language. The testing agency has decided to rank all the students (including both streams) as per their average marks. In case, there is a tie, then marks in physics for a science student and the marks in English for a arts student is given a preference to break the tie. Write a C program that will take *n* students as input and display the details of top three students.

Note: Your program must ensure that there is a single structure created for storing the details of student from different streams. You are free to use additional data members to the structure if needed. Try to minimize the memory space used to store the examination information.

[C05, 7 Marks, Apply]

Q6. Indian mobile number consists of 10 digits. The first four digits from the left represent the area code while the next 6 digits represent the extension. For example, 9910235674 is a valid mobile number, where 9910 represent the area code, while 235674 represent the extension. A file called **MobileNumberRecord** contains 100 such mobile numbers written using the following structure:

```
struct MobileNo
{
    int area;
    int extension;
};
```

Write a C program that will create a new file called **Filtered** that stores all those mobile numbers which are divisible by the sum of digits of area code.

For example, 8034125730 is divisible by the sum of digits of its area code ($8+0+3+4=15$)

[C04, 6 Marks, Apply]

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